Operator

作用于一个函数上得到1个新的函数.

基本运算性质.

(1) 算符相等.

$$\hat{F}u = \hat{G}u.$$

$$\hat{F} = \hat{G}$$

(2) 单位算符.

$$\hat{I}u = u$$

(3) 算符的和.

$$(\hat{F}+\hat{G})u = \hat{F}u + \hat{G}u$$

有交换律

$$\hat{F}+\hat{G} = \hat{G}+\hat{F}$$

结合律.

$$(\hat{F}+\hat{G})+\hat{M} = \hat{F}+(\hat{G}+\hat{M})$$

(4) 算符乘积.

$$(\hat{F}\hat{G})u = \hat{F}(\hat{G}u)$$

一般地, $\hat{F}\hat{G} \neq \hat{G}\hat{F}$

若 $\hat{F}\hat{G} = \hat{G}\hat{F}$, 称 $\hat{F}$、$\hat{G}$ 对易; 若 $\hat{F}\hat{G} = -\hat{G}\hat{F}$, 称 $\hat{F}$、$\hat{G}$ 反对易.

(5) 逆算符

$$\hat{F}\hat{G} = \hat{I}$$

(6) 算符的复共轭、转置与厄米共轭

线性代数中2函数的内积.

(a) 内积.

$$(u,v) \equiv \int u^* v \, d\tau = \left(\int u v^* d\tau\right)^* = (v,u)^*$$

性质:

$$(u,u) \geq 0$$

$$(u,v)^* = (v,u)$$

$$(u, c_1 v_1 + c_2 v_2) = c_1 (u, v_1) + c_2 (u, v_2)$$

$$(c_1 u_1 + c_2 u_2, v) = c_1^* (u_1, v) + c_2^* (u_2, v)$$

(b) 算符的复共轭算符 $\hat{F}^*$. 由 $\hat{F}$ 表达式中复量换为共轭复量构成.

例如:

$$\hat{p}_x = \frac{\hbar}{i}\frac{\partial}{\partial x} \qquad \hat{p}_x^* = -\frac{\hbar}{i}\frac{\partial}{\partial x} = -\hat{p}_x$$

(c) 转置算符.

定义: $\int d\tau \, u^* \tilde{\hat{F}} v \equiv \int d\tau \, v \hat{F} u^*$, 即

$$(u, \tilde{\hat{F}} v) = (v^*, \hat{F} u^*)$$

性质.

(i) $\frac{\tilde{\partial}}{\partial x} = -\frac{\partial}{\partial x}$

证明：

$$\int_{-\infty}^{\infty} dx\,(u^*\tilde{\frac{\partial}{\partial x}}v) = \int_{-\infty}^{\infty} dx\,(v\frac{\partial}{\partial x}u^*) = vu^*\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} dx\,(u^*\frac{\partial}{\partial x}v)$$

$$= -\int_{-\infty}^{\infty} dx\,u^*\frac{\partial}{\partial x}v$$

即，

$$\tilde{\frac{\partial}{\partial x}} = -\frac{\partial}{\partial x}.$$

上述推导使用了束缚态条件. 非束缚态？

(ii)

$$\tilde{p_x} = \frac{\hbar}{i}\tilde{\frac{\partial}{\partial x}} = -\frac{\hbar}{i}\frac{\partial}{\partial x} = -p_x.$$

(iii) $\widetilde{\hat{A}\hat{B}} = \tilde{\hat{B}}\tilde{\hat{A}}$

(d) 厄米共轭算符.

定义.

$$(u, \hat{F}^\dagger v) \equiv (\hat{F}u, v)$$

或

$$\int u^* \hat{F}^\dagger v\,d\tau \equiv \int (\hat{F}u)^* v\,d\tau$$

$$(u, \hat{F}^\dagger v) = (\hat{F}u, v) = (v, \hat{F}u)^* = (v^*, \hat{F}^* u^*) = (u, \tilde{\hat{F}}^* v)$$

$$\hat{F}^\dagger = \tilde{\hat{F}}^*$$

(7) 线性算符.

$$\hat{F}(c_1u_1 + c_2u_2) = c_1\hat{F}u_1 + c_2\hat{F}u_2$$

本征值方程、本征值、本征函数.

若算符 $\hat{F}$ 作用于一函数 ψ，等于 ψ 乘上一常数 λ.

$$\hat{F}\psi = \lambda\psi$$

则：上述方程称本征值方程，ψ 称本征函数，λ 称 $\hat{F}$ 的本征值.

量子力学中的力学量.

若量子力学中的力学量 F 在经典力学中有相应的力学量，则表示这个力学量的算符 $\hat{F}$ 由经典表示式 $F(r,p)$ 中将 p 换为算符 $\hat{p}$ 而得出。

经典		量子
$\vec{F}=\vec{F}(\vec{r},\vec{p})$	$\longrightarrow$	$\hat{F}=\hat{F}(\hat{r},\hat{p})=\hat{F}(r,-i\hbar\nabla)$

例：

动量算符：$\hat{\vec{p}} = -i\hbar\nabla$.

坐标算符：$\hat{\vec{r}} = \vec{r}$

角动量算符：$\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}} = \vec{r} \times (-i\hbar\nabla) = -i\hbar\vec{r}\times\nabla$.

哈密顿算符：$\hat{H} = \frac{\hat{p}^2}{2\mu} + U(\vec{r}) = -\frac{\hbar^2}{2\mu}\nabla^2 + U(\hat{\vec{r}})$.

算符与力学量关系：

若算符 $\hat{F}$ 表力学量 F，那么当体系处 $\hat{F}$ 的本征态 ψ 时，力学量 F 有确定值，该值为 $\hat{F}$ 在 ψ 中的本征值.

厄米算符 Hermite operator

若对 $\forall$ 2 函数 ψ, φ, 算符 $\hat{F}$ 满足:

$$\int \psi^* \hat{F} \varphi \, dx = \int (\hat{F}\psi)^* \varphi \, dx .$$

$\hat{F}$ 称厄米算符.

① 厄米算符本征值为实数.

② 量子力学的算符均为厄米算符.

设 $\hat{F}$ 的本征值方程:

$$\hat{F}\psi = \lambda \psi .$$

$$\int \psi^* (\hat{F}\phi) dx = \int (\hat{F}\psi)^* \phi \, dx$$

取 $\phi = \psi$.

$$\lambda \int \psi^* \psi \, dx = \int \psi^* (\lambda \psi) dx = \int \psi^* (\hat{F}\psi) dx$$

$$= \int (\hat{F}\psi)^* \psi \, dx = \int (\lambda \psi)^* \psi \, dx$$

$$= \lambda^* \int \psi^* \psi \, dx .$$

即 $\lambda = \lambda^*$

例: 动量算符.

$$\int_{-\infty}^{\infty} \psi^* \hat{p}_x \phi \, dx = \int_{-\infty}^{\infty} \psi^* \left(-i\hbar \frac{\partial}{\partial x}\right) \phi \, dx = -i\hbar \int_{-\infty}^{\infty} \psi^* \frac{\partial}{\partial x} \phi \, dx$$

$$\xrightarrow{\text{分部积分}} -i\hbar \psi^* \phi \Big|_{-\infty}^{\infty} + i\hbar \int_{-\infty}^{\infty} \phi \, d\psi^* = -i\hbar \psi^* \phi \Big|_{-\infty}^{\infty} + i\hbar \int_{-\infty}^{\infty} \phi \frac{\partial}{\partial x} \psi^* \, dx .$$

$$= \int_{-\infty}^{\infty} (\hat{p}_x \psi)^* \phi \, dx . \quad (\text{束缚态}: (\psi)_{x\to\pm\infty} = (\phi)_{x\to\pm\infty} = 0 .)$$

动量算符与角动量算符

动量算符

本征值方程：

$$\frac{\hbar}{i}\nabla\psi_p(\vec{r}) = \vec{p}\psi_p(\vec{r})$$

直角坐标中分量形式：

$$\begin{cases} \frac{\hbar}{i}\frac{\partial}{\partial x}\psi_p(r) = p_x\psi_p(r) \\ \frac{\hbar}{i}\frac{\partial}{\partial y}\psi_p(r) = p_y\psi_p(r) \\ \frac{\hbar}{i}\frac{\partial}{\partial z}\psi_p(r) = p_z\psi_p(r) \end{cases}$$

$$\Rightarrow \quad \psi_p(r) = C\exp\left(\frac{i}{\hbar}\vec{p}\cdot\vec{r}\right)$$

归一化

$$\phi(\vec{r}) = \int_{\vec{p}} b_{\vec{p}}\,\psi_{\vec{p}}\,d\vec{p}.$$

$$\phi^* = \int_{\vec{p}} b^*_{\vec{q}}\,\psi^*_{\vec{q}}\,d\vec{q}.$$

$$\psi_{\vec{p}} = Ce^{i\vec{p}\cdot\vec{r}/\hbar}.$$

$$\psi^*_{\vec{q}} = Ce^{-i\vec{q}\cdot\vec{r}/\hbar}$$

$$1 = \int_{\vec{r}}|\phi|^2 d\vec{r} = \int_{\vec{r}}\phi^*\phi\,d\vec{r} = \int_{\vec{r}}\int_{\vec{q}} b^*_{\vec{q}}\,\psi^*_{\vec{q}}\,d\vec{q}\int_{\vec{p}} b_{\vec{p}}\,\psi_{\vec{p}}\,d\vec{p}\,d\vec{r}$$

$$= \int_{\vec{q}} b^*_{\vec{q}}\int_{\vec{p}} b_{\vec{p}}\int_{\vec{r}}\psi^*_{\vec{q}}\psi_{\vec{p}}\,d\vec{r}\,d\vec{p}\,d\vec{q} = \int_{\vec{q}} b^*_{\vec{q}}\int_{\vec{p}} b_{\vec{p}}\,C^2(2\pi\hbar)^3\delta(\vec{p}-\vec{q})\,d\vec{p}\,d\vec{q}$$

$$\int_{\vec{r}}\psi^*_{\vec{q}}\psi_{\vec{p}}\,d\vec{r} = C^2(2\pi\hbar)^3\delta(\vec{p}-\vec{q})$$

$$= C^2(2\pi\hbar)^3 \int_{\vec{g}} b_{\vec{g}}^* b_{\vec{g}}\, d\vec{g} = C^2(2\pi\hbar)^3 \int_{\vec{g}} |b_{\vec{g}}|^2 d\vec{g} = C^2(2\pi\hbar)^3$$

$$\int \psi_{\vec{g}}^* \psi_{\vec{p}}\, d\vec{r} = \int C e^{-i\vec{g}\cdot\vec{r}/\hbar} C e^{i\vec{p}\cdot\vec{r}/\hbar}\, d\vec{r} = C^2 \int e^{i(\vec{p}-\vec{g})\cdot\vec{r}/\hbar}\, d\vec{r}$$

$$= C^2 \int e^{i[(p_x-g_x)x + (p_y-g_y)y + (p_z-g_z)z]/\hbar}\, dx\, dy\, dz.$$

$$= C^2 2\pi\hbar\,\delta(p_x-g_x)\, 2\pi\hbar\,\delta(p_y-g_y)\, 2\pi\hbar\,\delta(p_z-g_z)$$

$$= C^2(2\pi\hbar)^3 \delta(\vec{p}-\vec{g})$$

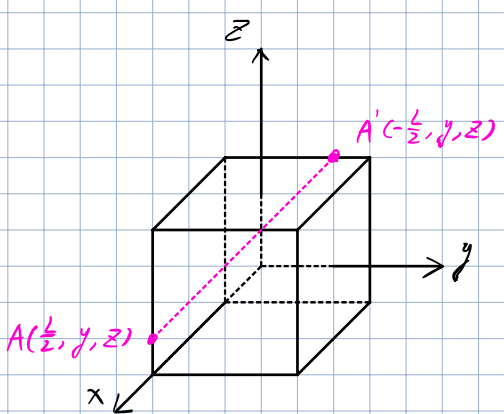
$$\Rightarrow \quad C = (2\pi\hbar)^{-\frac{3}{2}} \qquad \psi = (2\pi\hbar)^{-\frac{3}{2}} e^{i\vec{p}\cdot\vec{r}/\hbar}$$

箱归一化. 用箱分立本征值再令箱 $\to\infty$.

我们已知:

① 粒子限制于边长为 L 的正方体箱内.

② 周期性边界条件.



$$\psi_{\vec{p}}(A) = \psi_{\vec{p}}(A')$$

$$C e^{i(p_x \frac{L}{2} + p_y y + p_z z)/\hbar} = C e^{i(p_x(-\frac{L}{2}) + p_y y + p_z z)/\hbar}$$

$$e^{ip_x L/\hbar} = 1 \Rightarrow p_x L/\hbar = 2n_x\pi, \quad n_x = 0, \pm1, \pm2, \cdots.$$

$$p_x = 2n_x\pi\hbar/L \quad p_y = 2n_y\pi\hbar/L \quad p_z = 2n_z\pi\hbar/L$$

$$1 = \int_{L^3} |\psi_{\vec{p}}|^2 d\vec{r} = \int_{L^3} C^2 d\vec{r} = C^2 L^3 \Rightarrow C = L^{-\frac{3}{2}} \quad \psi_{\vec{p}} = L^{-\frac{3}{2}} e^{i\vec{p}\cdot\vec{r}/\hbar}$$

角动量算符

$$\hat{L} = \hat{r} \times \hat{p} = -i\hbar \vec{r} \times \nabla$$

直角坐标中三个分量：

$$\begin{cases} \hat{L}_x = y\hat{p}_x - z\hat{p}_y = \frac{\hbar}{i}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right) \\ \hat{L}_y = z\hat{p}_x - x\hat{p}_z = \frac{\hbar}{i}\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right) \\ \hat{L}_z = x\hat{p}_y - y\hat{p}_x = \frac{\hbar}{i}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right) \end{cases}$$

角动量平方算符：

$\vec{r} = \hat{e}_r r$

$\nabla \vec{F} = \hat{e}_r \frac{\partial F}{\partial r} + \hat{e}_\theta \frac{1}{r}\frac{\partial F}{\partial \theta} + \hat{e}_\varphi \frac{1}{r\sin\theta}\frac{\partial F}{\partial \varphi}$

$$\hat{L}^2\psi = \hat{\vec{L}} \cdot \hat{\vec{L}}\psi = (-i\hbar \vec{r} \times \nabla) \cdot (-i\hbar \vec{r} \times \nabla)\psi = -\hbar^2 (\vec{r} \times \nabla) \cdot (\vec{r} \times \nabla)\psi$$

$$= -\hbar^2 (\vec{r} \times \nabla) \cdot \left[\hat{e}_r r \times \left(\hat{e}_r \frac{\partial \psi}{\partial r} + \hat{e}_\theta \frac{1}{r}\frac{\partial \psi}{\partial \theta} + \hat{e}_\varphi \frac{1}{r\sin\theta}\frac{\partial \psi}{\partial \varphi}\right)\right]$$

$$= -\hbar^2 (\vec{r} \times \nabla) \cdot \left(\hat{e}_\varphi \frac{\partial \psi}{\partial \theta} - \hat{e}_\theta \frac{1}{\sin\theta}\frac{\partial \psi}{\partial \varphi}\right) \qquad (\vec{A} \times \vec{B}) \cdot \vec{C} = \vec{A} \cdot (\vec{B} \times \vec{C})$$

$$= -\hbar^2 \vec{r} \cdot \left[\nabla \times \left(\hat{e}_\varphi \frac{\partial \psi}{\partial \theta} - \hat{e}_\theta \frac{1}{\sin\theta}\frac{\partial \psi}{\partial \varphi}\right)\right]$$

$$\nabla \times \vec{F} = \frac{1}{r^2\sin\theta} \begin{vmatrix} \hat{e}_r & r\hat{e}_\theta & r\sin\theta\hat{e}_\varphi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \varphi} \\ F_r & rF_\theta & r\sin\theta F_\varphi \end{vmatrix}$$

$$= -\hbar^2 \vec{e}_r r \cdot \frac{1}{r^2\sin\theta} \begin{vmatrix} \hat{e}_r & r\hat{e}_\theta & r\sin\theta\hat{e}_\varphi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \varphi} \\ 0 & -r\frac{1}{\sin\theta}\frac{\partial \psi}{\partial \varphi} & r\sin\theta\frac{\partial \psi}{\partial \theta} \end{vmatrix}$$

$$= -\hbar^2 \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial \theta}\left(r\sin\theta\frac{\partial \psi}{\partial \theta}\right) + \frac{\partial}{\partial \varphi}\left(r\frac{1}{\sin\theta}\frac{\partial \psi}{\partial \varphi}\right)\right]$$

$$= -\hbar^2 \left[\frac{1}{\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial \psi}{\partial \theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 \psi}{\partial \varphi^2}\right]$$

$$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]$$

$x = r\sin\theta\cos\varphi \quad y = r\sin\theta\sin\varphi \quad z = r\cos\theta.$

$$\hat{L}_x = i\hbar\left(\sin\varphi\frac{\partial}{\partial\theta}+\cot\theta\cos\varphi\frac{\partial}{\partial\varphi}\right)$$

$$\hat{L}_y = -i\hbar\left(\cos\varphi\frac{\partial}{\partial\theta}-\cot\theta\sin\varphi\frac{\partial}{\partial\varphi}\right)$$

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial\varphi}.$$

$$\hat{L}_x^2 = -\hbar^2\left[\sin^2\varphi\frac{\partial^2}{\partial\theta^2}+2\cot\theta\sin\varphi\cos\varphi\frac{\partial^2}{\partial\theta\partial\varphi}+\cot^2\theta\cos^2\varphi\frac{\partial^2}{\partial\varphi^2}\right.$$

$$\left.+\cot\theta\cos^2\varphi\frac{\partial}{\partial\theta}-(\cot^2\theta+\csc^2\theta)\sin\varphi\cos\varphi\frac{\partial}{\partial\varphi}\right].$$

$$\hat{L}_y^2 = -\hbar^2\left[\cos^2\varphi\frac{\partial^2}{\partial\theta^2}-2\cot\theta\sin\varphi\cos\varphi\frac{\partial^2}{\partial\theta\partial\varphi}+\cot^2\theta\sin^2\varphi\frac{\partial^2}{\partial\varphi^2}\right.$$

$$\left.+\cot\theta\sin^2\varphi\frac{\partial}{\partial\theta}+(\cot^2\theta+\csc^2\theta)\sin\varphi\cos\varphi\frac{\partial}{\partial\varphi}\right]$$

$$\hat{L}_z^2 = -\hbar^2\frac{\partial^2}{\partial\varphi^2}$$

$$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]$$

① $\hat{L_z}$ 的本征方程与解.

$$\hat{L_z} e^{im\varphi} = L_z e^{im\varphi}.$$

$$L_z = m\hbar, \quad m = 0, 1, 2, \cdots.$$

② $\hat{L}^2$ 的本征值方程与球谐函数.

$$\hat{L}^2 Y_{l,m}(\theta, \varphi) = L^2 Y_{lm}(\theta, \varphi)$$

$$-\hbar^2 \left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi}\right] Y(\theta, \varphi) = L^2 Y(\theta, \varphi)$$

上式为 - Legendre 方程.

为使 $Y(\theta, \varphi)$ 在 θ 变化的闭域 $[0, \pi]$ 上有限. 有

$$L^2 = l(l+1)\hbar^2, \quad l = 0, 1, 2, \cdots.$$

$$m = -l, \cdots, l.$$

方程的解为球谐函数 $Y_{lm}(\theta, \varphi)$.

$$\begin{cases} Y_{lm}(\theta, \varphi) = (-1)^m N_{lm} P_l^m(\cos\theta) e^{im\varphi}, & m = 0, 1, \cdots, l, \\ Y_{lm}(\theta, \varphi) = (-1)^m Y^*_{l-m}(\theta, \varphi). & m = -1, -2, -3, \cdots, -l. \end{cases}$$

其中, $P_l^m(\cos\theta)$ 为 连带勒让德多项式, N_{lm} 为归一化因子.

由 $Y_{lm}(\theta, \varphi)$ 的正交归一化条件:

$$\int_0^{\pi}\int_0^{2\pi} Y^*_{lm}(\theta, \varphi) Y_{l'm'}(\theta, \varphi) \sin\theta \, d\theta \, d\varphi = \delta_{ll'}\delta_{mm'}, \quad m = 0, \pm1, \pm2, \cdots, \pm l.$$

$$N_{lm} = \sqrt{\frac{(l-m)!(2l+1)}{(l+m)!\,4\pi}}.$$

综上，

$\hat{L}^2$ 的本征值为 $l(l+1)\hbar^2$，所属本征函数为 $Y_{lm}(\theta,\varphi)$.

$$\hat{L}^2 Y_{lm}(\theta,\varphi) = l(l+1)\hbar^2 Y_{lm}(\theta,\varphi)$$

由于 l 表征角动量大小，称角量子数，m 则称磁量子数.

- 前几个球谐函数

$$Y_{00} = \frac{1}{\sqrt{4\pi}}$$

$$Y_{1\bar{1}} = \sqrt{\frac{3}{8\pi}}\sin\theta e^{-i\varphi},\quad Y_{10} = \sqrt{\frac{3}{4\pi}}\cos\theta,\quad Y_{11} = -\sqrt{\frac{3}{8\pi}}\sin\theta e^{i\varphi}$$

$$Y_{2\bar{2}} = \sqrt{\frac{15}{32\pi}}\sin^2\theta e^{-2i\varphi},\quad Y_{2\bar{1}} = \sqrt{\frac{15}{8\pi}}\sin\theta\cos\theta e^{-i\varphi},\quad Y_{20} = \sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$$

$$Y_{21} = -\sqrt{\frac{15}{8\pi}}\sin\theta\cos\theta e^{i\varphi},\quad Y_{22} = \sqrt{\frac{15}{32\pi}}\sin^2\theta e^{2i\varphi}$$

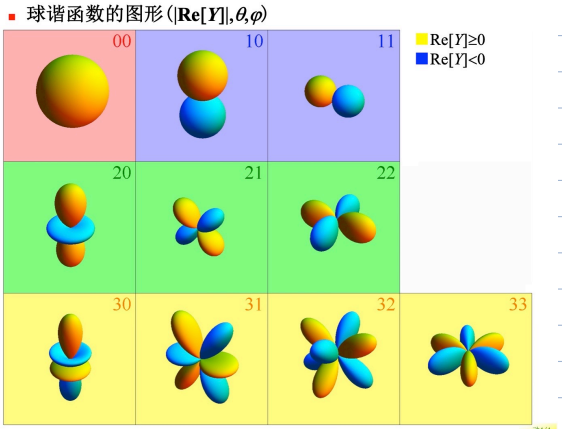
$$Y_{3\bar{3}} = \sqrt{\frac{35}{64\pi}}\sin^3\theta e^{-3i\varphi},\quad Y_{3\bar{2}} = \sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{-2i\varphi}$$

$$Y_{3\bar{1}} = \sqrt{\frac{21}{64\pi}}(5\cos^2\theta-1)\sin\theta e^{-i\varphi},\quad Y_{30} = \sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$$

$$Y_{31} = -\sqrt{\frac{21}{64\pi}}(5\cos^2\theta-1)\sin\theta e^{i\varphi},\quad Y_{32} = \sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{2i\varphi}$$

$$Y_{33} = -\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{3i\varphi}$$

- 球谐函数的图形 (|Re[Y]|,θ,φ)



电子在库仑场中运动

$-e,\mu$ ⊖

⊕ $+Ze$

类氢原子，如 $H(Z=1)$、$He^{+}(Z=2)$、$Li^{++}(Z=3)$

$$U=-\frac{Ze_s^2}{r}$$

定态薛定谔方程：

$$E\psi=\left(-\frac{\hbar^2}{2\mu}\nabla^2+U\right)\psi,$$

$$U=-\frac{Ze_s^2}{r}$$

分离变量，令 $\psi(r,\theta,\varphi)=R(r)Y(\theta,\varphi)$.

$$ERY=\left\{-\frac{\hbar^2}{2\mu r^2}\left[\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)+\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]-\frac{Ze_s^2}{r}\right\}(RY)$$

$$=-\frac{\hbar^2}{2\mu r^2}\left[Y\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{R}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right)+\frac{R}{\sin^2\theta}\frac{\partial^2 Y}{\partial\varphi^2}\right]-\frac{Ze_s^2}{r}(RY)$$

化为：

$$-\frac{2\mu r^2}{\hbar^2}E=\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{1}{Y}\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right)+\frac{1}{Y}\frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\varphi^2}+\frac{2\mu r^2}{\hbar^2}\frac{Ze_s^2}{r}$$

即：

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{2\mu r^2}{\hbar^2}\left(E+\frac{Ze_s^2}{r}\right)=-\left[\frac{1}{Y}\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right)+\frac{1}{Y}\frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\varphi^2}\right]=\lambda.$$

左边仅为 r 的函数，右边仅为 θ 与 φ 的函数，两边恒相等.

$$\begin{cases}-\left[\dfrac{1}{\sin\theta}\dfrac{\partial}{\partial\theta}\left(\sin\theta\dfrac{\partial Y}{\partial\theta}\right)+\dfrac{1}{\sin^2\theta}\dfrac{\partial^2 Y}{\partial\varphi^2}\right]=\lambda Y & \text{角向部分}\\ \dfrac{d}{dr}\left(r^2\dfrac{dR}{dr}\right)+\dfrac{2\mu r^2}{\hbar^2}\left(E+\dfrac{Ze_s^2}{r}\right)R=\lambda R & \text{径向部分}\end{cases}$$

1. 角向部分.

$$\lambda Y = -\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\varphi^2}\right]$$

$$\lambda\hbar^2 Y = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial}{\partial\varphi^2}\right]Y = \hat{L}^2 Y.$$

上式为电子角动量平方的本征值方程.

$$l(l+1)\hbar^2 Y_{lm} = \hat{L}^2 Y_{lm}.$$

$$\lambda = l(l+1).$$

$$l = 0, 1, 2, \cdots; \quad m = -l, \cdots, l.$$

2. 径向部分.

将角向部分 λ 值代入, 有:

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{2\mu}{\hbar^2}\left(E+\frac{Ze^2}{r}\right)R = \frac{\lambda}{r^2}R = \frac{l(l+1)}{r^2}R.$$

$$\Longleftrightarrow \quad \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left[\frac{2\mu}{\hbar^2}\left(E+\frac{Ze^2}{r}\right)-\frac{l(l+1)}{r^2}\right]R = 0.$$

(1) $E>0$.

对 E 的任何值, 方程均有满足波函数条件的解, 体系的能量具连续谱.

此时电子被电离, 可离开核运动至无限远处.

(2) $E<0$.

束缚态, 电子具分立谱.

① 无量纲化.

$$0=\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left[\frac{2\mu}{\hbar^2}\left(\frac{Ze_s^2}{r}+E\right)-\frac{l(l+1)}{r^2}\right]R,\quad R(0)\neq\infty\quad R(\infty)\neq\infty.$$

考虑

$$\frac{d^2(rR)}{dr^2}=\frac{d}{dr}\left(R+r\frac{dR}{dr}\right)=2\frac{dR}{dr}+r\frac{d^2R}{dr^2}$$

$$\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)=2r\frac{dR}{dr}+r^2\frac{d^2R}{dr^2}=r\frac{d^2(rR)}{dr^2}=r\frac{d^2u}{dr^2}$$

于是上式可化为:

$$0=\frac{du}{dr^2}+\left[\frac{2\mu}{\hbar^2}\left(\frac{Ze_s^2}{r}-|E|\right)-\frac{l(l+1)}{r^2}\right]\mu,$$

$$\mu(0)=0$$

$$\lim_{r\to\infty}\frac{\mu}{r}\neq\infty.$$

观察上式，试图通过变量替换，将上式化为:

$$\begin{cases}0=\dfrac{d^2\mu}{d\rho^2}+\left[\dfrac{\beta}{\rho}-\dfrac{1}{4}-\dfrac{l(l+1)}{\rho^2}\right]\mu.\\ \mu(0)=0\\ \lim\limits_{\rho\to\infty}\dfrac{\mu}{\rho}\neq\infty\end{cases}$$

比较后，有:

$$\begin{cases}\beta=\dfrac{2\mu Ze_s^2}{\alpha\hbar^2}=\dfrac{Ze_s^2}{\hbar^2}\left(\dfrac{\mu}{2|E|}\right)^{\frac{1}{2}}\\ \alpha=\left(\dfrac{8\mu|E|}{\hbar^2}\right)^{\frac{1}{2}}\\ \rho\equiv\alpha r\end{cases}$$

② 考虑 μ 渐近行为.

$$0=\frac{d^2\mu}{d\rho^2}+\left[\frac{\beta}{\rho}-\frac{1}{4}-\frac{l(l+1)}{\rho^2}\right]\mu,\quad\lim_{\rho\to\infty}\frac{u}{\rho}\neq\infty,\quad u(0)=0$$

$i\ \rho\to\infty.\quad u''-\frac{u}{4}=0\quad u=Ae^{-\frac{\rho}{2}}+Be^{\frac{\rho}{2}}=Ae^{-\frac{\rho}{2}}\propto e^{-\frac{\rho}{2}}$

ii $\rho \to 0$. $u'' - \frac{l(l+1)u}{\rho^2} = 0$.

考虑，

$$u = \rho^k$$

$$u' = k\rho^{k-1}$$

$$u'' = k(k-1)\rho^{k-2}$$

代入有：

$$0 = k(k-1)\rho^{k-2} - \frac{l(l+1)\rho^k}{\rho^2} = [k(k-1) - l(l+1)]\rho^{k-2}$$

$$0 = k^2 - k - l(l+1) \implies k = l+1,\ -l$$

$$u = A\rho^{l+1} + B\rho^{-l} \propto \rho^{l+1},\quad B = 0.$$

③求解 $u(\rho)$.

$$0 = \frac{d^2u}{d\rho^2} + \left[\frac{\beta}{\rho} - \frac{1}{4} - \frac{l(l+1)}{\rho^2}\right]u, \qquad \lim_{\rho\to\infty}\frac{u}{\rho} \neq \infty$$

由上，

$$u(\infty) \propto e^{-\frac{\rho}{2}} \qquad u(0) \propto \rho^{l+1}$$

令

$$u(\rho) = \rho^{l+1}e^{-\frac{\rho}{2}}f(\rho)$$

代入，有：

$$\rho f'' + (2l+2-\rho)f' - (l+1-\beta)f = 0$$

上式为合流超几何方程，$z\frac{d^2y}{dx^2} + (c-z)\frac{dy}{dz} - ay = 0$

解为：

$$f(\rho) = F(l+1-\beta,\ 2l+2,\ \rho) \xrightarrow[\rho\to\infty]{} e^{\rho} = e^{\alpha r}$$

$$u(r) = Ne^{-\frac{1}{2}\alpha r}r^{l+1}F(l+1-\beta,\ 2l+2,\ \rho) \xrightarrow[r\to\infty]{} Ne^{\frac{1}{2}\alpha r}r^{l+1} \xrightarrow[r\to\infty]{} \infty.$$

注意，$\rho \to \infty$ 时 $u \to \infty$. ∴要使合流超几何级数截断为合流超几何多项式，方程才满足波函数有界条件.

∴ $$l+1-\beta = -n_r \quad , \quad n_r = 0, 1, 2, \cdots.$$

记 $$\beta = n_r + l + 1 \equiv n, \quad n = 1, 2, \cdots. \quad l = 0, 1, 2, \cdots, n-1$$

$$u = \rho^{l+1} e^{-\frac{\rho}{2}} F(l+1-n, 2l+2, \rho)$$

④本征能量与本征函数.

本征能量

$$\beta = \frac{Ze_s^2}{\hbar^2}\left(\frac{\mu}{2|E|}\right)^{\frac{1}{2}} = n \Rightarrow E = -\frac{\mu Z^2 e_s^4}{2\hbar^2 n^2}$$

$$\alpha = \left(\frac{8\mu E}{\hbar^2}\right)^{\frac{1}{2}} = \left(\frac{8\mu}{\hbar^2}\frac{\mu Z^2 e_s^4}{2\hbar^2 n^2}\right)^{\frac{1}{2}} = \frac{2Z\mu e_s^2}{n\hbar^2} = \frac{2Z}{na_0}$$

其中， $$a_0 \equiv \frac{\hbar^2}{\mu e_s^2}$$

本征函数

$$R \propto \frac{u}{\rho} = \rho^l e^{-\frac{\rho}{2}} F(l+1-n, 2l+2, \rho)$$

$$R_{nl}(r) = N_{nl} \alpha^l r^l e^{-\frac{\alpha r}{2}} F(l+1-n, 2l+2, \alpha r)$$

由归一化条件：

$$\int_0^\infty R_{nl}^2 r^2 dr = 1$$

$$N_{nl} = \frac{2}{n^2(2l+1)!}\left[\frac{(n+l)!\, Z^3}{(n-l-1)!\, a_0^3}\right]^{\frac{1}{2}}.$$

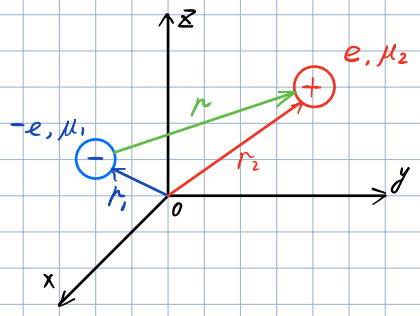
- 简并：同一个能级下有多个状态，状态的数目称简并度.

E_n；$\psi_{nlm}(r,\theta,\varphi)$；$n=1,2,\cdots$；$l=0,1,2,\cdots,n-1$；$m=-l,\cdots,l$

$$D_n=\sum_{l=0}^{n-1}\sum_{m=-l}^{l}1=n^2$$

对特定 n，组合 (l,m) 有 n^2 种取值可能，即简并度为 n^2

氢原子



一个电子与一个氢核之间的库仑相互作用.

假设氢核不固定，两体问题.

电子与氢核组成体系的薛定谔方程：

$$i\hbar\frac{\partial}{\partial t}\Psi(x_1,y_1,z_1;x_2,y_2,z_2;t)$$
$$=\left[-\frac{\hbar^2}{2m_1}\left(\frac{\partial^2}{\partial x_1^2}+\frac{\partial^2}{\partial y_1^2}+\frac{\partial^2}{\partial z_1^2}\right)-\frac{\hbar^2}{2m_2}\left(\frac{\partial^2}{\partial x_2^2}+\frac{\partial^2}{\partial y_2^2}+\frac{\partial^2}{\partial z_2^2}\right)+U\right]\Psi(x_1,y_1,z_1;x_2,y_2,z_2;t)$$

坐标换为两粒子相对坐标：

$$x=x_1-x_2,\quad y=y_1-y_2,\quad z=z_1-z_2$$

以 X, Y, Z 表体系的质心坐标

$$mX=m_1x_1+m_2x_2\qquad mY=m_1y_1+m_2y_2\qquad mZ=m_1z_1+m_2z_2$$

其中，$m=m_1+m_2$.

将粒子坐标微商变为相对坐标与质心坐标微商.

$$\frac{\partial}{\partial x_1}=\frac{\partial X}{\partial x_1}\frac{\partial}{\partial X}+\frac{\partial x}{\partial x_1}\frac{\partial}{\partial x}=\frac{m_1}{m}\frac{\partial}{\partial X}+\frac{\partial}{\partial x}$$

$$\frac{\partial^2}{\partial x_1^2}=\left(\frac{m_1}{m}\frac{\partial}{\partial X}+\frac{\partial}{\partial x}\right)^2=\frac{m_1^2}{m^2}\frac{\partial^2}{\partial X^2}+\frac{2m_1}{m}\frac{\partial^2}{\partial X\partial x}+\frac{\partial^2}{\partial x^2}$$

$$\frac{\partial}{\partial x_2}=\frac{\partial X}{\partial x_2}\frac{\partial}{\partial X}+\frac{\partial x}{\partial x_2}\frac{\partial}{\partial x}=\frac{m_2}{m}\frac{\partial}{\partial X}+\frac{\partial}{\partial x}$$

$$\frac{\partial^2}{\partial x_2^2}=\left(\frac{m_2}{m}\frac{\partial}{\partial X}+\frac{\partial}{\partial x}\right)^2=\frac{m_2^2}{m^2}\frac{\partial^2}{\partial X^2}+\frac{2m_2}{m}\frac{\partial^2}{\partial X\partial x}+\frac{\partial^2}{\partial x^2}$$

$\frac{\partial^2}{\partial y_1^2}$、$\frac{\partial}{\partial y_1}$、$\frac{\partial}{\partial z_1}$、$\frac{\partial}{\partial z_1^2}$ 同理. 代入后，有：

$$i\hbar\frac{\partial\Psi}{\partial t} = -\left[\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial X^2}+\frac{\partial^2}{\partial Y^2}+\frac{\partial^2}{\partial Z^2}\right)+\frac{\hbar^2}{2\mu}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)+U(x,y,z)\right]\Psi.$$

其中，$\mu=\frac{m_1m_2}{m_1+m_2}$ 为约化质量.

分离变量. 设 $\Psi=\psi(x,y,z)\phi(X,Y,Z)\chi(t)$

$$\psi\phi i\hbar\frac{d\chi}{dt} = -\phi\chi\frac{\hbar^2}{2\mu}\nabla_r^2\psi-\psi\chi\frac{\hbar^2}{2m}\nabla_R^2\phi+U\psi\phi\chi$$

$$\frac{i\hbar^2}{\chi}\frac{d\chi}{dt} = -\frac{1}{\psi}\frac{\hbar^2}{2\mu}\nabla_r^2\psi-\frac{1}{\phi}\frac{\hbar^2}{2m}\nabla_R^2\phi+U=E_{总}$$

左边的解：

$$\chi(t)=(e^{-\frac{i}{\hbar}E_{总}t}$$

右边化为：

$$-\frac{\hbar^2}{2m}\frac{1}{\phi}\left(\frac{\partial^2}{\partial X^2}+\frac{\partial^2}{\partial Y^2}+\frac{\partial^2}{\partial Z^2}\right)\phi-\frac{\hbar^2}{2\mu}\left(\frac{\partial^2}{\partial x}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\psi+U(x,y,z)=E_{总}$$

左边第一项只与 X, Y, Z 有关，第二，三项只与 x, y, z 有关. ∴ 它们分别等于常数且常数之和为 $E_{总}$.

$$-\frac{\hbar^2}{2\mu}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\psi+U(x,y,z)\psi=E\psi$$

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\phi=(E_{总}-E)\phi.$$

第二个方程为描写质心运动状态的波函数 ϕ 的方程. 且为能量为 $E_{总}-E$ 的自由粒子的定态薛定谔方程.

而在氢原子问题中，我们主要关注电子相对于核的运动状态，E 即为电子的能级. 于是有：

本征能量.

$$E_n = -\frac{\mu e_s^4}{2\hbar^2 n^2} = -\frac{e_s^2}{2a_\mu}\frac{1}{n^2}, \quad n=1,2,3,\cdots.$$

其中, $a_\mu = a_0$ 为玻尔半径.

- 电离能.

$$E_\infty - E_1 = 0 + \frac{\mu e_s^4}{2\hbar^2 1^2} = \frac{\mu c^2 e_s^4}{2(\hbar c)^2} = \frac{0.511\times10^6\times14.4}{2\times1973^2} \approx 13.6\,eV.$$

- 光谱:

$$\nu = \frac{|E_n - E_m|}{2\pi\hbar} = \frac{m_e e_s^4}{4\pi\hbar^3}\left(\frac{1}{n'^2} - \frac{1}{n^2}\right) = R_\infty c\left(\frac{1}{n'^2} - \frac{1}{n^2}\right)$$

其中, $R_\infty = 10973731.568527\ m^{-1}$ 为里德伯常量.

- 径向概率密度与角向概率密度.

$$1 = \int R_{nl}^2(r) r^2 dr \Longrightarrow W_{nl}(r) = R_{nl}^2(r)\,dr$$

$$1 = \iint |Y_{lm}(\theta,\varphi)|^2 \sin\theta\, d\theta\, d\varphi = \iint |Y_{lm}(\theta,\varphi)|^2 d\Omega$$

$$W_{lm}(\Omega) = |Y_{lm}(\theta,\varphi)|^2 = \frac{(l-|m|)!(2l+1)}{(l+|m|)!2\pi}\left[P_l^m(\cos\theta)\right]^2$$

厄米算符本征函数的正交性.

正交：若函数 ϕ_1 与 ϕ_2 满足

$$\int \phi_1^* \phi_2 \, d\tau = 0$$

定理：属于不同本征值的2个本征函数相互正交.

证明：设 (λ_k, ϕ_k) 与 (λ_l, ϕ_l) 为 $\hat{F}$ 的本征值与本征函数. 且 $\lambda_k \neq \lambda_l$.

$$\hat{F}\phi_k = \lambda_k \phi_k \Rightarrow (\hat{F}\phi_k)^* = (\lambda_k \phi_k)^* = \lambda_k^* \phi_k^* = \lambda_k \phi_k^*$$

$$\Rightarrow (\hat{F}\phi_k)^* \phi_l = \lambda_k \phi_k^* \phi_l \Rightarrow \int (\hat{F}\phi_k)^* \phi_l \, d\tau = \int \lambda_k \phi_k^* \phi_l \, d\tau = \lambda_k \int \phi_k^* \phi_l \, d\tau$$

$$\hat{F}\phi_l = \lambda_l \phi_l \Rightarrow \phi_k^* (\hat{F}\phi_l) = \phi_k^* (\lambda_l \phi_l) \Rightarrow \int \phi_k^* (\hat{F}\phi_l) \, d\tau = \int \phi_k^* (\lambda_l \phi_l) \, d\tau = \lambda_l \int \phi_k^* \phi_l \, d\tau.$$

$\hat{F}$ 为厄米算符.

$$\int (\hat{F}\phi_k)^* \phi_l \, d\tau = \int \phi_k^* (\hat{F}\phi_l) \, d\tau$$

$$\Rightarrow \lambda_k \int \phi_k^* \phi_l \, d\tau = \lambda_l \int \phi_k^* \phi_l \, d\tau.$$

$$(\lambda_k - \lambda_l) \int \phi_k^* \phi_l \, d\tau = 0$$

$$\int \phi_k^* \phi_l \, d\tau = 0$$

正交归一.

① 本征函数构本征值组成分立谱.

若本征函数 ϕ_k 已归一，即

$$\int \phi_k \phi_k^* \, d\tau = 1.$$

又∵

$$\int \phi_k^* \phi_l \, d\tau = 0$$

有:

$$\int \phi_k^* \phi_l \, d\tau = \delta_{kl} = \begin{cases} 1 & (k=l) \\ 0 & (k \neq l) \end{cases}$$

② 本征函数构本征值组成有简并时分立谱.

采用施密特方法.

将 $\hat{F}$ 对属于同一个本征值 λ_n 的多个本征函数 ϕ_{ni} 重新线性组合成一组归一的，相互正交的新函数 ψ_{nj}

i 线性组合

$$\hat{F}\phi_{ni} = \lambda_n \phi_{ni} \quad , \quad i = 1, 2, \cdots, f.$$

$$\psi_{nj} = \sum_{i=1}^{f} A_{ji} \phi_{ni} \quad , \quad j = 1, 2, \cdots, f.$$

ii 利用正交归一关系确定 A_{ij}

$$\int \psi_{nj}^* \psi_{nj'} \, d\tau = \sum_{i,i'=1}^{f} A_{ji}^* A_{j'i'} \int \phi_{ni}^* \phi_{ni'} \, d\tau = \delta_{jj'}$$

③本征函数的本征值组成连续谱.

归一化为 δ 函数.

$$\int \phi_\lambda^* \phi_{\lambda'} \, d\tau = \delta(\lambda - \lambda')$$

一些正交归一函数

(1) 线性谐振子的能量本征函数

$$\psi_n(x) = N_n e^{-\frac{1}{2}\alpha^2 x^2} H_n(\alpha x)$$

组成正交归一系

$$N_n N_n' \int_{-\infty}^{\infty} e^{-\frac{1}{2}\alpha^2 x^2} H_n(\alpha x) H_{n'}(\alpha x) \, dx = \delta_{nn'}$$

(2) 角动量算符 $\hat{L}_z$ 的本征函数.

$$\Phi_m(\varphi) = \frac{1}{\sqrt{2\pi}} e^{im\varphi} \qquad (m = 0, \pm 1, \pm 2, \cdots)$$

组成归一正交系

$$\int_0^{2\pi} \Phi_m^*(\varphi) \Phi_{m'}(\varphi) \, d\varphi = \delta_{mm'}$$

角动量算符 $\hat{L}^2$ 属于本征值 $l(l+1)\hbar^2$ 的本征函数

$$Y_{lm}(\theta, \varphi) = N_{lm} P_l^m(\cos\theta) e^{im\varphi}.$$

组成正交归一系

$$\int_0^{\pi} \int_0^{2\pi} Y_{lm}^*(\theta, \varphi) Y_{l'm}(\theta, \varphi) \sin\theta \, d\theta \, d\varphi = \delta_{ll'}$$

由上式可得连带勒让德多项式的正交性.

$$2\pi N_{lm} N_{l'm} \int_0^{\pi} P_l^m(\cos\theta) P_{l'}^m(\cos\theta) \sin\theta \, d\theta = \delta_{ll'}$$

合并:

$$\int_0^{\pi}\int_0^{2\pi} Y_{lm}^*(\theta,\varphi) Y_{l'm'}(\theta,\varphi) \sin\theta \, d\theta d\varphi = \delta_{ll'}\delta_{mm'}$$

算符与力学量关系

在前面我们引入了一个基本假设：

若算符 $\hat{F}$ 表力学量 F，那么当体系处 $\hat{F}$ 的本征态 ϕ 时，力学量 F 有确定值，该值为 $\hat{F}$ 在 ϕ 中的本征值.

该假设无法完全解决算符与力学量的关系这一问题.

我们要问：

若体系不处于 $\hat{F}$ 的本征态，而处于任意一个态，此时算符 $\hat{F}$ 与它所表示的力学量有何关系？

1. 本征函数完全性.

若 $\hat{F}$ 为满足一定条件的厄米算符，其正交归一本征函数为 ϕ_n，对应本征值为 λ_n.

则：某一函数 $\psi_n(x)$ 可按 ϕ_n 展开为级数

$$\psi(x) = \sum_n c_n \phi_n(x).$$

这种性质称本征函数的完全性.

由 ϕ_n 正交归一性：

$$\int \phi_m^*(x)\,\psi(x)\,dx = \sum_n c_n \int \phi_m^*(x)\,\phi_n(x)\,dx$$

$$= \sum_n c_n \delta_{mn} = c_m.$$

即

$$c_n = \int \phi_n^*(x)\,\psi(x)\,dx.$$

若 $\psi(x)$ 即体系状态波函数且已归一，有：

$$\int \psi^*(x)\,\psi(x)\,dx = \sum_{mn} c_m^* c_n \int \phi_m^*(x)\,\phi_n(x)\,dx$$

$$= \sum_{mn} c_m^* c_n \delta_{mn} = \sum_n |c_n|^2 = 1.$$

又由前文，测量力学量 F 必有 $F=\lambda_i$，λ_i 对应一 c_i

$|c_n|^2$ 具有概率的意义。c_n 常被称为概率振幅.

我们有以下一个关于力学量与算符的一个基本假定：

量子力学中表示力学量的算符都为厄米算符，它们的本征函数组成完全系。当体系处于波函数 $\psi(x)$ 所描写的状态时，测得力学量 F 所得的值必定为算符 $\hat{F}$ 的本征值之一，测得 λ_n 的概率为 $|c_n|^2$

力学量 F 在 ψ 态的期望

分立谱：

$$\bar{F} = \sum_n \lambda_n |c_n|^2 = \int \psi^*(x)\,\hat{F}\,\psi\,dx$$

连续谱 / 分立谱 + 连续谱：

$$\bar{F} = \sum_n \lambda_n |c_n|^2 + \int \lambda |c_\lambda|^2 d\lambda.$$

例：氢原子处于基态时电子动量的概率分布.

氢原子处于基态 $\psi_{100}(\vec{r})$，动量本征态 $\psi_{\vec{p}}(\vec{r})$

将氢原子基态波函数 ψ_{100} 按动量算符本征函数 $\psi_{\vec{p}}$ 展开.

$$\psi_{100} = \int c_{\vec{p}}\,\psi_{\vec{p}}(\vec{r})\,d\vec{p}$$

$$c_{\vec{p}} = \int \psi_{\vec{p}}^{*}(\vec{r})\,\psi_{100}(\vec{r})\,d\tau$$

将

$$\psi_{100}(\vec{r}) = \frac{1}{\sqrt{\pi a_0^3}} e^{-\frac{r}{a_0}} \qquad \psi_{\vec{p}}^{*}(\vec{r}) = \frac{1}{(2\pi\hbar)^{\frac{3}{2}}} e^{-\frac{i}{\hbar}\vec{p}\cdot\vec{r}}$$

$$c_{\vec{p}} = \frac{1}{\pi^2 (2a_0\hbar)^{\frac{3}{2}}} \int_0^{r}\int_{-1}^{1}\int_0^{2\pi} e^{-\frac{r}{a_0}} e^{-\frac{i}{\hbar}pr\cos\theta}\, r^2\, dr\, d\cos\theta\, d\varphi.$$

$$= \frac{(2a_0\hbar)^{\frac{3}{2}}\hbar}{\pi\left[a_0^2p^2 + \hbar^2\right]^2}$$

上式仅与 p 大小有关，与方向无关，可得动量概率密度.

$$|c_{\vec{p}}|^2 = \frac{8a_0^3\hbar^5}{\pi^2(a_0^2p^2+\hbar^2)^2}$$

氢原子处于基态时，电子动量的绝对值在 $p \longrightarrow p+dp$ 范围内的概率等于 $|c_{\vec{p}}|^2$ × 动量空间的体积元 $4\pi p^2 dp$. 即

$$\omega(p)\,dp = \frac{32}{\pi}\left(\frac{\hbar}{a_0}\right)^5 \frac{p^2 dp}{\left[\left(\frac{\hbar}{a_0}\right)^2 + p^2\right]^4}$$

算符间对易关系

以坐标算符 $\hat{x}$ 与动量算符 $\hat{p}_x$ 为例.

$$\hat{x}\hat{p}_x\psi \equiv \hat{x}(\hat{p}_x\psi) = \hat{x}\left(-i\hbar\frac{\partial}{\partial x}\psi\right) = x\left(-i\hbar\frac{\partial}{\partial x}\psi\right)$$

$$= -i\hbar x\frac{\partial\psi}{\partial x}$$

$$\hat{p}_x\hat{x}\psi = -i\hbar\frac{\partial}{\partial x}(x\psi) = -i\hbar\psi - i\hbar x\frac{\partial}{\partial x}\psi$$

$$(\hat{x}\hat{p}_x - \hat{p}_x\hat{x})\psi = i\hbar\psi.$$

$\therefore$ dim

$$[\hat{x}, \hat{p}_x] = \hat{x}\hat{p}_x - \hat{p}_x\hat{x} = i\hbar$$

同理我们还有:

$$\begin{cases} [\hat{\alpha}, \hat{p}_\beta] = i\hbar\delta_{\alpha,\beta} \\ [\hat{p}_\alpha, \hat{p}_\beta] = 0 \end{cases} \qquad (\alpha, \beta = x, y, z)$$

由上面例子发现:

1. 两算符作用于同一函数, 结果次序对作用的顺序.

2. 定义 对易.

if $[\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = 0$ then call $\hat{x}$、$\hat{p}$ 对易.

if $[\hat{x}, \hat{p}] \neq 0$ then call 不对易.

3. 动量算符分量与对应的坐标算符分量是不对易的.

4. 动量算符各分量间为对易的.

角动量算符的对易关系:

$$[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z$$

$$[\hat{L}_y, \hat{L}_z] = i\hbar \hat{L}_x$$

$$[\hat{L}_z, \hat{L}_x] = i\hbar \hat{L}_y$$

$$[\hat{L}_x, \hat{L}^2] = [\hat{L}_y, \hat{L}^2] = [\hat{L}_z, \hat{L}^2] = 0$$

定理:

$[\hat{F}, \hat{G}] = 0 \iff \hat{F}, \hat{G}$ 有共同本征函数且组成完全系.

证:

$$[\hat{F}, \hat{G}] = 0 \iff \begin{cases} \hat{F}\phi_n = \lambda_n \phi_n \\ \hat{G}\phi_n = \mu_n \phi_n \end{cases} \iff (\hat{F}\hat{G} - \hat{G}\hat{F})\phi_n = 0.$$

设 ψ 为任意波函数, 按 ϕ 展开为级数.

$$\psi = \sum_n \phi_n$$

$$(\hat{F}\hat{G} - \hat{G}\hat{F})\psi = 0$$

$$\hat{F}\hat{G} - \hat{G}\hat{F} = 0$$

对易与力学量的完全集合

力学量的完全集合：

一组完全确定体系状态的力学量.

集合中，力学量数目一般等于体系自由度.

不对易与测不准关系

设 $\hat{F}$ 与 $\hat{G}$ 对易关系

$$[\hat{F},\hat{G}]=i\hat{k}$$

$\bar{F},\bar{G},\bar{k}$ 表 $\hat{F},\hat{G},\hat{k}$ 在态 ψ 中期望.

$$\Delta\hat{F}=\hat{F}-\bar{F}.\quad \Delta\hat{G}=\hat{G}-\bar{G}$$

考虑积分：

$$I(\xi)=\int|(\xi\Delta\hat{F}-i\Delta\hat{G})\psi|^2d\tau\geqslant 0.$$

展开：

$$I(\xi)=\int(\xi\Delta\hat{F}\psi-i\Delta\hat{G}\psi)[\xi(\Delta\hat{F}\psi)^*+i(\Delta\hat{G}\psi)^*]d\tau$$

$$=\xi^2\int(\Delta\hat{F}\psi)(\Delta\hat{F}\psi)^*d\tau-i\xi\int[(\Delta\hat{G}\psi)(\Delta\hat{F}\psi)^*-(\Delta\hat{F}\psi)(\Delta\hat{G}\psi)^*]d\tau$$

$$+\int(\Delta\hat{G}\psi)(\Delta\hat{G}\psi)^*d\tau$$

注意到，$\Delta\hat{F}$、$\Delta\hat{G}$ 均为 Hermite 算符.

$$\int\psi^*\Delta\hat{F}\psi\,d\tau=\int(\Delta\hat{F}\psi)^*\psi\,d\tau.\quad ,\ \cdots$$

$$I(\xi) = \xi^2 \int \psi^* (\Delta\hat{F})^2 \psi \, d\tau - i\xi \int \psi^* (\Delta\hat{F}\Delta\hat{G} - \Delta\hat{G}\Delta\hat{F}) \psi \, d\tau + \int \psi^* (\Delta\hat{G})^2 \psi \, d\tau$$

$$= \xi^2 \int \psi^* (\Delta\hat{F})^2 \psi \, d\tau + \xi \int \psi^* \hat{k} \psi \, d\tau + \int \psi^* (\Delta\hat{G})^2 \psi \, d\tau$$

$$= \overline{(\Delta\hat{F})^2} \xi^2 + \bar{k}\xi + \overline{(\Delta\hat{G})^2} \geq 0$$

由代数中二次式理论可知：

$$\bar{k}^2 - 4 \overline{(\Delta\hat{F})^2} \, \overline{(\Delta\hat{G})^2} \leq 0$$

即

$$\overline{(\Delta\hat{F})^2} \, \overline{(\Delta\hat{G})^2} \geq \frac{\bar{k}^2}{4}$$

若 $\bar{k} \neq 0$，$\Delta\hat{F}$，$\Delta\hat{G}$ 的均方偏差不同时为 0. 上关系称不确定关系.

例如，对坐标算符与动量算符，

$$\overline{(\Delta x)^2} \, \overline{(\Delta p)^2} \geq \frac{\hbar^2}{4}$$

- 不确定关系为量子力学中基本关系，反映了微观粒子波粒二象性.

论论：如何理解势垒贯穿中，在势垒内部粒子动能为负值？

由于坐标算符与动量算符不对易，势能算符与动能算符不对易. 势能能动能不能同时有确定值.

当粒子在势垒范围内被发现时，根据不确定关系，粒子动能在某一范围内不确定.

∵ 势垒宽度为 a.

$$\Delta x \leq a$$

由不确定关系：

$$\overline{(\Delta x)^2}\ \overline{(\Delta p_x)^2} \geq \frac{\hbar^2}{4}$$

$$\overline{(\Delta p_x)^2} \geq \frac{\hbar^2}{4a^2}$$

∴ 粒子动能的最小不确定范围：

$$\Delta T = \frac{\overline{(\Delta p_x)^2}}{2m} \geq \frac{\hbar^2}{8ma^2}$$

线性谐振子的零点能也可用不确定关系说明：

振子的平均能量：

$$\bar{E} = \frac{\bar{p^2}}{2m} + \frac{1}{2}m\omega^2\bar{x^2}$$

坐标期望值：

$$\bar{x} = N_n^2 \int_{-\infty}^{\infty} e^{-\alpha^2x^2} H_n^2(\alpha x)\, x\, dx = 0$$

$e^{-\alpha^2x^2}H_n^2(\alpha x)$ 为 x 的偶函数

动量期望值：

$$\bar{p} = \frac{\hbar}{i} N_n^2 \int_{-\infty}^{\infty} e^{-\frac{\alpha^2x^2}{2}} H_n(\alpha x) \frac{d}{dx}\left[e^{-\frac{\alpha^2x^2}{2}} H_n(\alpha x)\right] dx$$

$$= -\bar{p}$$

$$\bar{p} = 0$$

由均方偏差公式：

$$\overline{(\Delta F)^2} = \overline{(F-\bar{F})^2} = \overline{F^2} - 2\bar{F}\bar{F} + \bar{F}^2 = \overline{F^2} - \bar{F}^2$$

$$\overline{(\Delta x)^2} = \overline{x^2} \qquad \overline{(\Delta p)^2} = \overline{p^2}$$

$$\bar{E} = \frac{\overline{(\Delta p)^2}}{2m} + \frac{1}{2}m\omega^2\overline{(\Delta x)^2}$$

由不确定关系：

$$\overline{(\Delta p)^2} \geqslant \frac{\hbar^2}{4\overline{(\Delta x)^2}}$$

代入，有：

$$\bar{E} \geqslant \frac{\hbar^2}{8m\overline{(\Delta x)^2}} + \frac{1}{2}m\omega^2\overline{(\Delta x)^2} \geqslant \frac{1}{2}\hbar\omega.$$

力学量期望随时间的变化　守恒定律

守恒定律

波函数 $\Psi(x,t)$ 所描写的态中，$\hat{F}$ 期望值为：

$$\overline{F(t)} = \int \Psi^*(x,t)\hat{F}\Psi(x,t)\,dx$$

讨论 $\overline{F}$ 随时间变化，对时间求微商：

$$\frac{d\overline{F}}{dt} = \int \psi^* \frac{\partial \hat{F}}{\partial t}\psi\,dx + \int \frac{\partial \Psi^*}{\partial t}\hat{F}\Psi\,dx + \int \Psi^*\hat{F}\frac{\partial \Psi}{\partial t}dx.$$

由薛定谔方程：

$$\frac{\partial \Psi}{\partial t} = \frac{1}{i\hbar}\hat{H}\Psi.$$

$$\frac{\partial \Psi^*}{\partial t} = -\frac{1}{i\hbar}(\hat{H}\Psi)^*$$

$$\int(\hat{H}\Psi)^*\hat{F}\Psi\,dx = \int \Psi^*\hat{H}\hat{F}\Psi\,dx$$

代入：

$$\frac{d\overline{F}}{dt} = \int \Psi^*\frac{\partial \hat{F}}{\partial t}\Psi\,dx + \frac{1}{i\hbar}\int \Psi^*\hat{F}\hat{H}\Psi\,dx - \frac{1}{i\hbar}\int(\hat{H}\Psi)^*\hat{F}\Psi\,dx$$

$$= \int \Psi^*\frac{\partial \hat{F}}{\partial t}\Psi\,dx + \frac{1}{i\hbar}\int \Psi^*(\hat{F}\hat{H}-\hat{H}\hat{F})\Psi^*\,dx$$

即

$$\frac{d\overline{F}}{dt} = \overline{\frac{\partial F}{\partial t}} + \frac{1}{i\hbar}\overline{(\hat{F}\hat{H}-\hat{H}\hat{F})}$$

$$= \overline{\frac{\partial F}{\partial t}} + \frac{1}{i\hbar}\overline{[\hat{F},\hat{H}]}$$

若 $\hat{F}$ 不显含时间，

$$\frac{\partial \hat{F}}{\partial t} = 0.$$

$$\frac{d\overline{F}}{dt} = \frac{1}{i\hbar}\overline{[\hat{F},\hat{H}]}$$

若 $\hat{F}$ 还与 $\hat{H}$ 对易，

$$\frac{d\hat{F}}{dt} = 0.$$

称 F 为运动守恒量.

举例：

(1) 自由粒子的动量.

$$\hat{H} = \frac{1}{2m}\hat{p}^2$$

$\therefore$

$$\frac{d\bar{p}}{dt} = \frac{1}{i\hbar}\overline{[\hat{p}, \hat{H}]} = 0$$

(2) 中心力场中运动粒子的角动量.

$$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]$$

$$\hat{H} = -\frac{\hbar^2}{2\mu r^2}\left[\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right] + U(r)$$

$$= -\frac{\hbar^2}{2\mu r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\hat{L}^2}{2\mu r^2} + U(r)$$

$$[\hat{L}, \hat{L}^2] = [\hat{L}^2, \hat{L}^2] = 0, \quad \hat{L}(\theta, \varphi), \ \hat{L}^2(\theta, \varphi)$$

$$[\hat{L}, \hat{H}] = [\hat{L}^2, \hat{H}] = 0$$

$$\frac{d\bar{L}}{dt} = \frac{d\bar{L}^2}{dt} = 0$$

粒子在中心力场中运动时，角动量平方和角动量分量为运动恒量.

(3) 能量守恒定律.

哈密顿算符不显含时间 t 又与自身对易.

$$\frac{\partial\hat{H}}{\partial t} = 0 \qquad [\hat{H}, \hat{H}] = 0 \quad \Longrightarrow \quad \frac{d\bar{H}}{dt} = 0$$

当体系哈密顿量不显含时间时，能量为运动恒量.

14) 宇称守恒定律.

宇称算符 $\hat{P}$：改变函数坐标变量的符号，也称空间反演.

$$\hat{P}\Psi(x,t) = \Psi(-x,t)$$

$$\hat{P}^2\Psi(x,t) = \hat{P}(\hat{P}\Psi(x,t)) = \Psi(x,t)$$

- 宇称算符的本征值与本征函数.

$$\hat{P}\Psi_1 = \Psi_1 \qquad \hat{P}\Psi_2 = -\Psi_2$$

称属于1的 Ψ_1 具有偶宇称，属于-1的 Ψ_2 具有奇宇称.

- 宇称算符不显含时间 t.

若哈密顿算符在空间反演后不变，

$$\hat{H}(x) = \hat{H}(-x) \qquad [\hat{P},\hat{H}] = 0$$

$$\Rightarrow \qquad \frac{d\bar{P}}{dt} = 0$$

宇称是运动恒量。体系的能量本征函数具确定的宇称.

中心力场

力场作用线总通过固定一点.

例如：库仑场、各向同性谐振子场、无限深球方势阱.

线性中心势与对数中心势.

哈密顿算符

$$\hat{H}=-\frac{\hbar^2}{2\mu r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)+\frac{\hat{L}^2}{2\mu r^2}+U(r)$$

角动量守恒

$$[\hat{L},\hat{H}]=[\hat{L}^2,\hat{H}]=0$$

系统有3个自由度，用角动量3个分量表示.

用三个对易守恒量的共同本征态对定态进行分类.

$$(\hat{H},\hat{L}^2,\hat{L}_z)\longrightarrow(n,l,m)$$

径向方程：

$$\psi_{l,m}(r,\theta,\varphi)=R_l(r)Y_{l,m}(\theta,\varphi)$$

$$\left[\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d}{dr}\right)+\frac{2\mu}{\hbar^2}(E-U)-\frac{l(l+1)}{r^2}\right]R_l=0$$

- 径向动能
- 离心势能

由于 $m=-l,\cdots,l$，能量本征值至少 $2l+1$ 重简并.

求解径向方程：

令 $R_l(r) = \frac{u_l(r)}{r}$

$$u \equiv rR_l$$

$$\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) = r\frac{d^2u_l}{dr^2}$$

$$0 = \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR_l}{dr}\right) + \left[\frac{2\mu}{\hbar^2}(E-U) - \frac{l(l+1)}{r^2}\right]R_l$$

$$= \frac{1}{r^2}r\frac{d^2u}{dr^2} + \left[\frac{2\mu}{\hbar^2}(E-U) - \frac{l(l+1)}{r^2}\right]\frac{u_l}{r}$$

$$0 = \frac{d^2u_l}{dr^2} + \left[\frac{2\mu}{\hbar^2}(E-U) - \frac{l(l+1)}{r^2}\right]u_l$$

非束缚态：E是连续的.

束缚态：E为量子化的，出现径向量子数.

称 $l = 0, 1, 2, 3, \cdots$ 的态依次为 s, p, d, f, $\cdots$ 态.

$r \to 0$, $R_l \propto r^l$

无限深球方势阱

$$U(r)=\begin{cases}0 & r<a\\ \infty & r>a\end{cases}$$

径向方程：

$$0=\left[\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d}{dr}\right)+\frac{2\mu}{\hbar^2}(E-U)-\frac{l(l+1)}{r^2}\right]R_l$$

$$=\left[\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d}{dr}\right)+\frac{2\mu E}{\hbar^2}-\frac{l(l+1)}{r^2}\right]R_l.$$

其中，$R_l(0)$，$R_l(a)\neq\infty$

1. $l=0$ 的态.

令 $u\equiv rR_l$

$$\left(\frac{d^2}{dr^2}+\frac{2\mu E}{\hbar^2}\right)u_0=u_0''+k^2u_0=0$$

$$k^2=\frac{2\mu E}{\hbar^2}$$

$$R_l(0),\ R_l(a)\neq\infty \Longrightarrow u_0(0)=u_0(a)=0$$

$$u_0=A\sin kr+B\cos kr$$

$$\left.\begin{aligned}u_0(0)&=B=0\\ u_0(a)&=A\sin ka=0\end{aligned}\right\}\Longrightarrow \quad ka=n_r\pi.$$

$$k^2a^2=n_r^2\pi^2=\frac{2\mu E}{\hbar^2}a^2 \Longrightarrow E_{n_r,0}=\frac{n_r^2\pi^2\hbar^2}{2\mu a^2} \qquad n_r=1,2,\cdots$$

归一化：$A=\sqrt{\frac{2}{a}}$.　　$R_{n_r,0}=\sqrt{\frac{2}{a}}\,r\sin\frac{n_r\pi r}{a}$　　$n_r=1,2,\cdots$

- 能级与波函数与一维无限深方势阱相同.

2. $l \neq 0$ 的态.

令 $\rho \equiv kr$ $\quad k^2 = \frac{2\mu E}{\hbar^2}$

$$\left[\frac{k^3}{\rho^2}\frac{d}{d\rho}\left(\frac{\rho^2}{k}\frac{d}{d\rho}\right) + k^2 - \frac{k^2 l(l+1)}{\rho^2}\right] R_l = 0$$

又 $R_l(\rho) = \frac{u_l(\rho)}{\sqrt{\rho}}$，化为：

$$\left[\frac{2}{\rho}\frac{d}{d\rho} + \frac{d^2}{d\rho^2} + 1 - \frac{l(l+1)}{\rho^2}\right](u_l \rho^{-\frac{1}{2}}) = 0$$

$$u_l'' + \frac{1}{\rho}u_l' + \left[1 - \frac{(l+\frac{1}{2})^2}{\rho^2}\right] u_l = 0$$

$$R_l(a) = u_l(a)\, a^{-\frac{1}{2}} = 0 \Longrightarrow u_l(a) = 0$$

本征方程为 1 个 $l+\frac{1}{2}$ 阶的 Bessel 方程：

$$u_l(\rho) = A J_{l+\frac{1}{2}}(\rho) + \underbrace{B J_{-l-\frac{1}{2}}(\rho)}_{=0} = A J_{l+\frac{1}{2}}(\rho)$$

$$R_l(r) = \frac{A J_{l+\frac{1}{2}}(kr)}{\sqrt{kr}}$$

$$A J_{l+\frac{1}{2}}(ka) = 0 \Longrightarrow ka = x_{n_r, l} \quad n_r = 0, 1, \cdots$$

$$\frac{2\mu E}{\hbar^2} a^2 = k^2 a^2 = x_{n_r, l}^2 \Longrightarrow E_{n_r l} = \frac{x_{n_r l}^2 \hbar^2}{2\mu a^2}$$

- 能级 $E_{n_r, l}$ 依赖于角量子数 l 与径向量子数 n_r，能量随 n_r 与 l 增大而增大.
- 当 $a \to \infty$，$r \to \infty$ 时，$J_{l+\frac{1}{2}} \to 0$ 能量不可连续变化. 为自由粒子.